MATH256
Linear Algebra HW#4

1. (2.8.8) $\boldsymbol{v_1} = \begin{bmatrix} -2 \\ 0 \\ 6 \end{bmatrix}$, $\boldsymbol{v_2} = \begin{bmatrix} -2 \\ 3 \\ 3 \end{bmatrix}$, $\boldsymbol{v_3} = \begin{bmatrix} 0 \\ -5 \\ 5 \end{bmatrix}$ and $\mathbf{p} = \begin{bmatrix} -6 \\ 1 \\ 17 \end{bmatrix}$.
Determine if $\mathbf{p}$ is in Col A , where A =$[\boldsymbol{v_1} \ , \ \boldsymbol{v_2} \ , \ \boldsymbol{v_3}]$.

2. (2.8.10) With $\mathbf{u} = \begin{bmatrix} -5 \\ 5 \\ 3 \end{bmatrix}$ and A as in question 1 , determine if $\mathbf{u}$ is in Nul A .

In Question 3 and 4 give integers p and q such that Nul A is a subspace of $\mathbb{R}^p$ and Col A is a sunspace of $\mathbb{R}^q$

3. (2.8.11)

$$\begin{bmatrix} 3 & 2 & 1 & -5 \\ -9 & -4 & 1 & 7 \\ 9 & 2 & -5 & 1 \end{bmatrix}$$

NulA = row number?
ColA = column number?
Are these always true?

4. (2.8.12)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ -5 & -1 & 0 \\ 2 & 7 & 11 \\ 3 & 3 & 4 \end{bmatrix}$$

In Question 5 and 6 Display a matrix A and an echelon form of A. Find a basis for Col A and basis for Nul A .

5. (2.8.24)

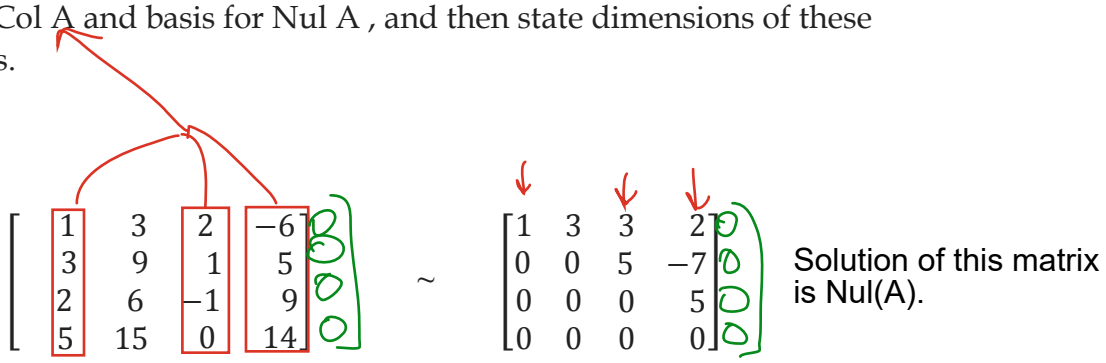
$$\begin{bmatrix} 3 & -6 & 9 & 0 \\ 2 & -4 & 7 & 2 \\ 3 & -6 & 6 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 5 & 4 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

6. (2.8.26)

$$\begin{bmatrix} 3 & -1 & -3 & 1 & 8 \\ 3 & 1 & 3 & 0 & 2 \\ 0 & 3 & 9 & -1 & -4 \\ 6 & 3 & 9 & -2 & 6 \end{bmatrix} \sim \begin{bmatrix} 3 & -1 & -3 & 0 & 6 \\ 0 & 2 & 6 & 0 & -4 \\ 0 & 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In Question 7 and 8 Display a matrix A and an echelon form of A. Find a basis for Col A and basis for Nul A , and then state dimensions of these subspaces.

7. (2.9.9)



$$\begin{bmatrix} 1 & 3 & 2 & -6 \\ 3 & 9 & 1 & 5 \\ 2 & 6 & -1 & 9 \\ 5 & 15 & 0 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution of this matrix is Nul(A).

8. (2.9.10)

$$\begin{bmatrix} 1 & -2 & -1 & 5 & 4 \\ 2 & -1 & 1 & 5 & 6 \\ -2 & 0 & -2 & 1 & -6 \\ 3 & 1 & 4 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$